

25-9-2024

STATS lec: no 01 Thursday

Introduction lecture:

(سنت لیا)

Book:

Introduction to Statistical theory By
Sher Muhammad Chaudhary (Part-I)

Study of facts and figures.

Definition of Statistics:

"It is the branch of science which is based on five major components i.e collection, tabulation, presentation/classification, analysis & interpretation."

Characteristics:

we use it as single / plural.

deal with quantable values, also deal with probability. it is based on historical facts & figures.

"quantitative data."

Limitations:

prove on average.

Collection:

"الطائرا"

not gone through any statistical procedure
raw data also called primary data.

6 types

if gone through statistical measure
secondary data.

1. Direct personal investigation

- (third party)
- i) Indirect personal investigation.
- ii) questioners.
- iii) telephonic calls / social media.
- iv)

23-9-2024

STATS lec no 02

Friday

Chapter No: 01

Introduction

Characteristics of Statistics:

whole aggregates
don't deal with individual value. all
formulae deal with variables.

- it is unit free. all data unit must be same.
- mutually exclusive, nothing common.

Branches of Statistics:

Descriptive & Inferential

↓
Summarization of data, collection,
presentation, classifi, analysis

Inferential:

"It is branch of statistics in which
we draw conclusions about population
on basis of sample data".

Population:

- living or non living population
- finite or infinite

when pop becomes too large, on
analysis becomes diff.

in inferential, we do conclusion on sample data.

Population Size:

It is denoted by N

Sample size:

It is denoted by n

~~discrete~~

→ Importance:

→ Observations and variables:

Variables denoted by X, Y, Z , etc
constant a, k, n , etc

Types:

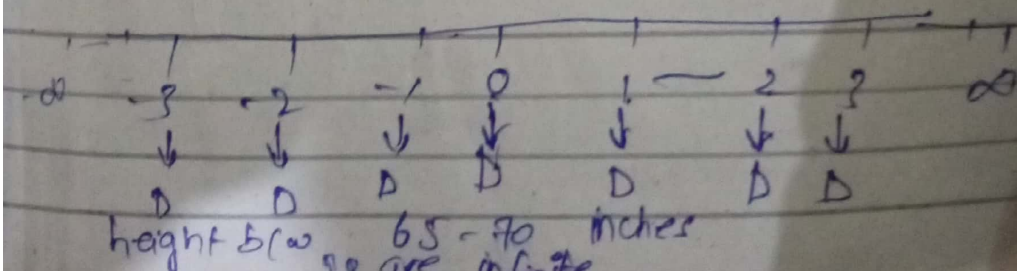
Discrete & Continuous variable

① Discrete: (countable)

A variable which cannot assume any particular value b/w an interval.

Continuous:

A variable which can assume any particular value b/w an interval.



Qualitative: A variable which deals non-numeric values also known as attribute (to express quality).

X	45	46	47	48	no of flowers
No of trees	3	8	4	1	

Quantitative variable: A variable which deals numeric values

→ Measurement Scale: "4 types of Scales"

(i) Nominal scale:

- mutually exclusive: nothing common
- used in forms.
- comes from nominees. they nominate.

(ii) Ordinal or Ranking:

assign Ranks

ex: ascending / descending

(iii) Interval & Ratio:

not true zero

temperature,

ht. data

true zero point

match pitch

Age, weight

• Error of Measurement:

absolute epsilon ϵ .

capture the small error occurs due to fluctuation.

• Significant Digits:

Collection of Data:

two types of data

Primary Data (raw data)

the data that does not go through any statistical treatment.

Secondary (classified)

the data that does go through any statistical treatment.

Primary has different methods:

(iv) Enumerator:

"سوال پوچھ کر"

"census (example)"

"we do questioning."

سوال پوچھ کر

responded

(v) local sources:

used in primary & secondary data.

Secondary:

Experimental Research:

Editing: my leading / inconsistent do
no mistake in grading procedure
by mistake.

inconsistent → total marks are 50
but you write 58

misuse: No knowledge of statistics
that is misuse. (Chapter one ended)

2-10-2024

STAT lec no 03 Wednesday

Presentation of Data:

Classification:

two types / ways:

classification / Diagrams

↓
mutually exclusive qualitative categories

Marks	No of students
0-4	3
5-9	12
10-14	14

don't have similarity in 0-4 & 5-9

classification can be qualitative / quantitative.

	Gender		Contingency table
	Male	Female	
Pass	40	60	
Fail	80	20	

Aims of classification:

- ① produce large data in ^{understood} summary.

Tabulation:

Types of table:

2 types

① General purpose table

② Specific purpose table

(i) have 2 or more things

(ii) have inf of one thing.

Specific purpose if

Name	Reg No	OS
------	--------	----

Name	Reg No	OS	Stats	Eng
------	--------	----	-------	-----

General purpose
✓

Table parts:

Row caption = stub

column caption = Boxhead

title: table name (Marks)

Prefatory notes: from how much, additional thing. (total marks)

units: of how many decimal point in marks, for reg, unit is kg.

Stub: ~~column~~ row name

Body of source notes - reference of secondary data

Ex: 2.1 Do it

$$\begin{aligned}\text{total pop} &= R + U \\ T.P &= 271,076 \\ U &= 46,740\end{aligned}$$

$$271,076 - 46,740 = R$$

$$R = M + F$$

$$M = \text{Mar} + U.M$$

$$F = \text{Mar} + U.F$$

Continuous freq. distribution

• Frequency Distribution: (Grouped Data)

Primary Data : Total = 25

Quantitative Data

20, 21, 25, 17, 23, 24, 20, 24, 23, 25, 18, 20, 9,
24, 13, 20, 5, 6, 12, 17, 15, 19, 07, 09, 10, 8, 3, 22,
20, 13, 19. (ungrouped Data)

Parts of freq Distribution:

Marks	No. of Students
0-4	
5-9	
10-14	
15-19	
20-24	
25-29	

inclusive : where value end add the
exclusive : where gp is include th

if ind0
0-5 add 5 here
if exc
5-9 add 5 here

• Constructing freq Distribution

~~1) Range~~ ① Range: (R)

$$R = X_m - X_0 = 25 - 3 = 22$$

where $X_m = \text{max value} = 25$
 $X_0 = \text{min value} = 3$

② No of classes / groups: (m)

$$m = 1 + 3.3 \log(N) \quad N = 31$$

$$= 1 + 3.3 \log(31) = 1 + 3.3(1.4914)$$

$$m = 1 + 4.922$$

$$m = 5.922$$

$$m \approx 6$$

③ Size of class / class width / height / interval (h):

$$h = \frac{R}{m} = \frac{22}{6} = 3.67$$

$$h = 4$$

Marks	Tally	No. of students (f)
3-6	34	03
7-10		12
11-14		1
15-18	I	
19-22	III	
23-26	IIII	
		34

Major Points:

for inclusive write same value

for exclusive //

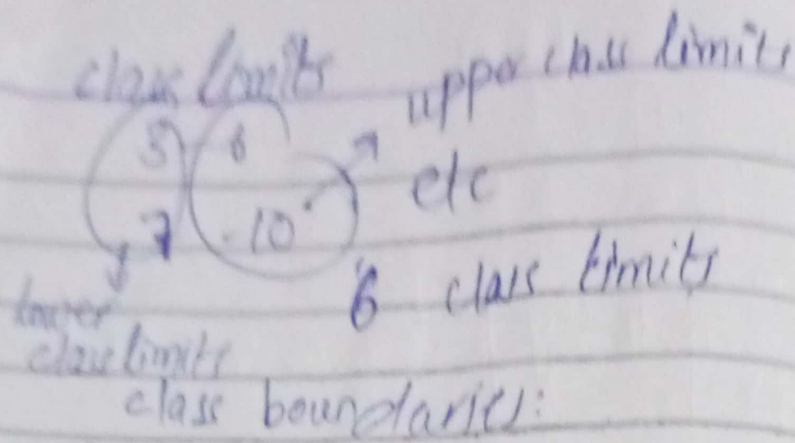
if not specify write one less Ao

if inclusive write where group end

if exclusive // // // start

for 6 write (3-6)

for 6 write (7-10)



Marks	Class Boundaries	Class Mark / class midpoint (x)
3 - 6	2.5 - 6.5	4.5
7 - 10	6.5 - 10.5	8.5
11 - 14	10.5 - 14.5	12.5
15 - 18	14.5 - 18.5	16.5
19 - 22	18.5 - 22.5	20.5
23 - 26	22.5 - 26.5	24.5

in some cases: U.L = upper limit
 L.L = lower limit
 arrow value must be same

$$C.B = \frac{U.L + L.L}{2} = \frac{7 + 6}{2} = 6.5$$

0.5 minus

minus from Lower limit
 add from upper limit

Marks	C.B	
3 - 6.5	2.5 - 6.5	= 0.25
7 - 10.5	6.75 - 10.75	
11 - 14.5	10.75 - 14.75	
15 - 18.5	14.75 - 18.75	
19 - 22.5	18.75 - 22.75	
23 - 26.5	22.75 - 26.75	

$$\text{Mid point} = \frac{\text{Upper Bound} + \text{Lower Bound}}{2}$$

9-10-2024

STAT 101 1004

Wednesday

• Stem & leaf Display:

29, 40, 98, 57, 69, 48, 18, 18,
37, 18, 79, 7, 37, 11, 48, 333,
28, 78, 39, 208, 86, 23, 6969,
2005, 999, 2003, 9999, 3-14, 9.27,
1.27, 6.63.

to arrange some, we use stem & leaf
ex: write in ascending order

29
↓ stem (leading digit)
leaf (trailing digit) 07 = 7

Stem (leading digit)	Leaf (trailing digit)	Ordered values
0	7	7
1	5 8 5 1 .27	0.27, 1.27, 5.27, 5.8
2	9 8 08 3 005 003	3, 8, 9, 98, 908
3	7 7 33 .14	.14, 7, 7, 33
4	0 8 8	0, 8, 8
5	2 9	2, 9
6	9 969 .63	.63, 9, 969
7	9 6	6, 9
8	3	3
9	8 99 999	8, 9, 999

Arranged Data:-

1.27, 3.14, 6.63, 7, 11, 15, 15, 18, 23, 28, 29,
~~2003, 2005~~ 37, 37, 40, 48, 48, 52, 59,
69, 76, 79, 83, 98, 99, 208, 333, 999,
2002, 2005, 6969

→ two types of representation,

• Diagrams (Diagrammatical representation)
Presentation of statistical data by
means of geometrical figures is
called diagrammatical representation.

• Graphical Representation
Presentation of statistical data by
means of geometrical curve is
called graphical representation.

→ Diagrams:

one dimensional

two //

three 4

Circle // (π)

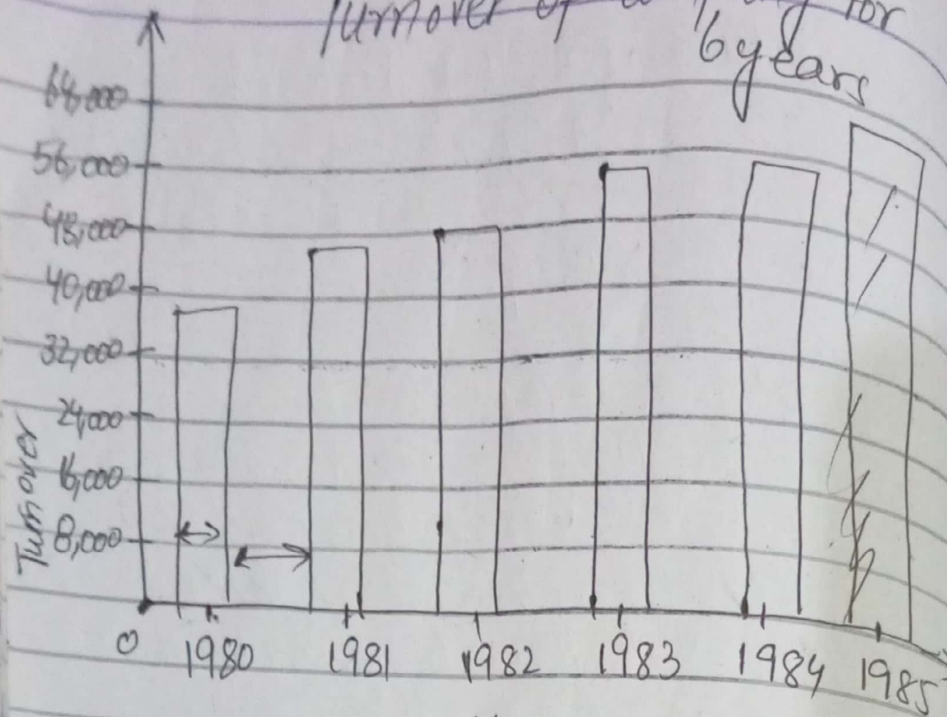
cartogram (map)

pictogram: made with help of
images & symbols.

→ Simple Bar Diagram: (one dimensional)
* single characteristic (quality)

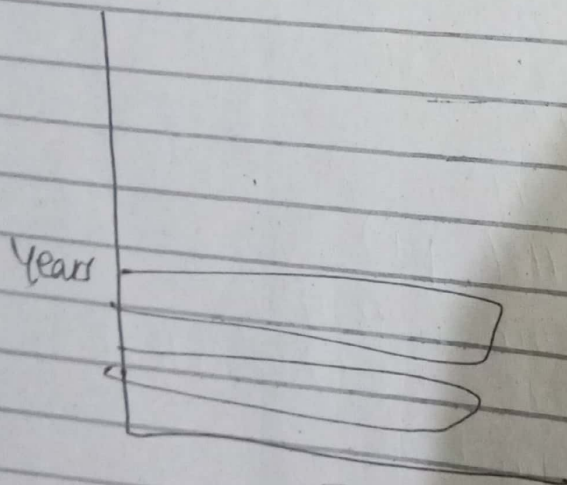
Example:

Turnover of company for 6 years



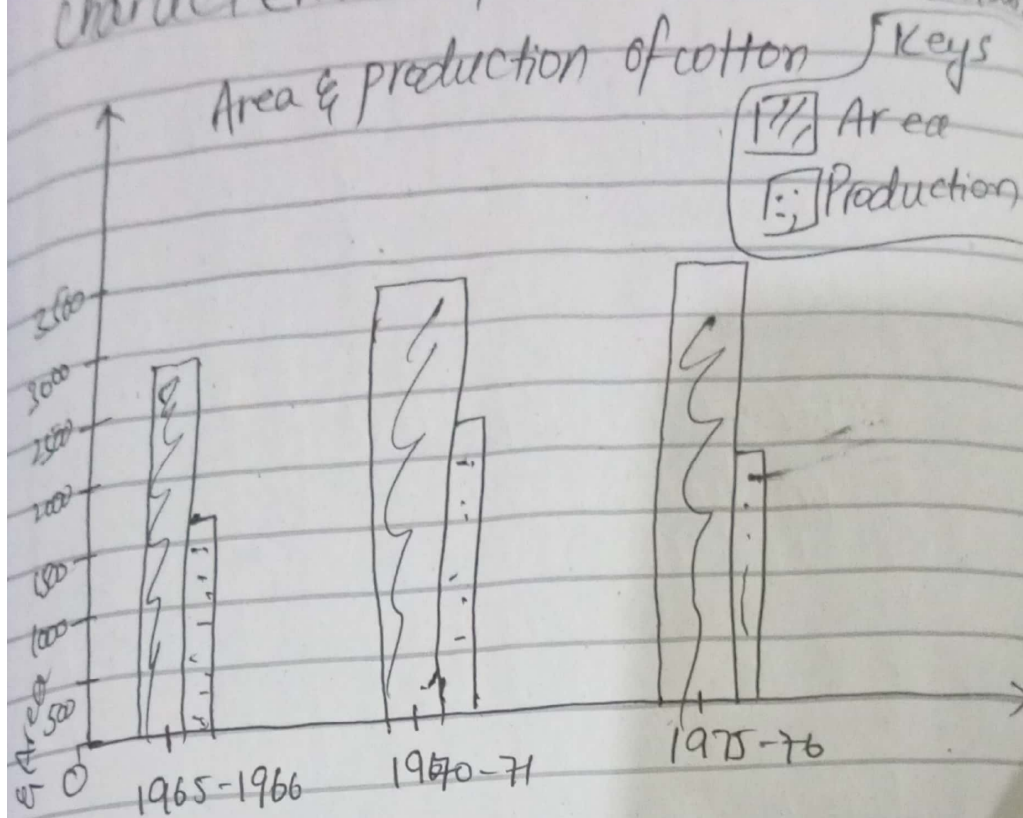
bar distance must be same

- write label for diagram
- diag name is vertical Bar Diag.



Horizontal Bar Diagram

Multiple Bar chart Diagram:
 characteristic of two or more than two



example:

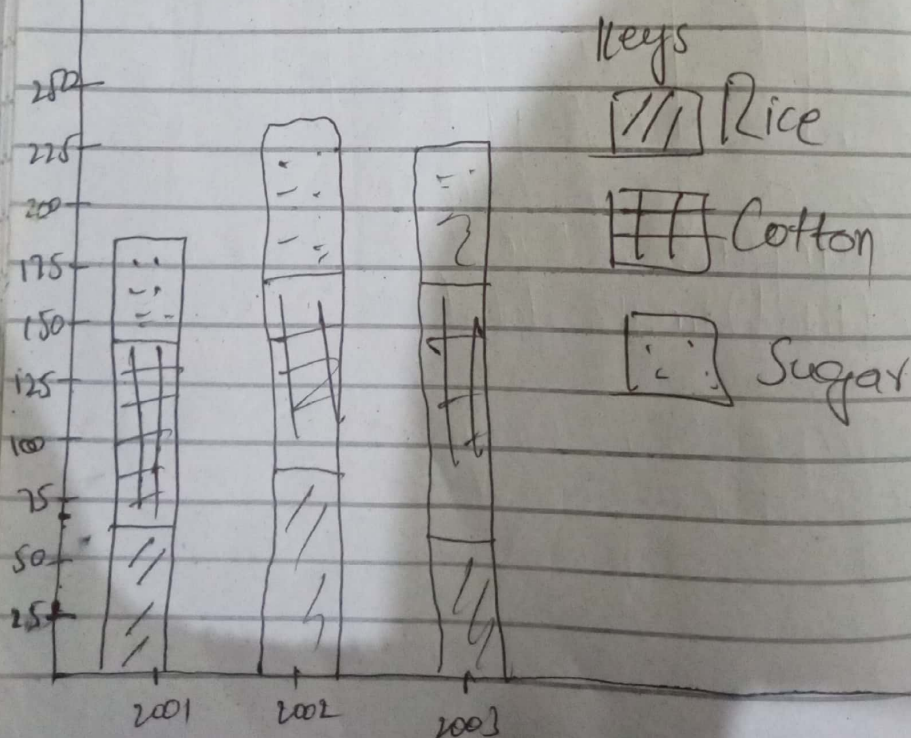
	Production		
	Rice	Cotton	Sugar
Year			
2001			
2002			
2003			

Component Bar Chart:
 (total) make total

Year	Rice (tons)	Production		Total
		Cotton	Sugar	
2001	72	68	40	180
2002	82	90	55	227
2003	60	110	55	225

Graph on full page Bar of total

Component Bar Chart Showing production

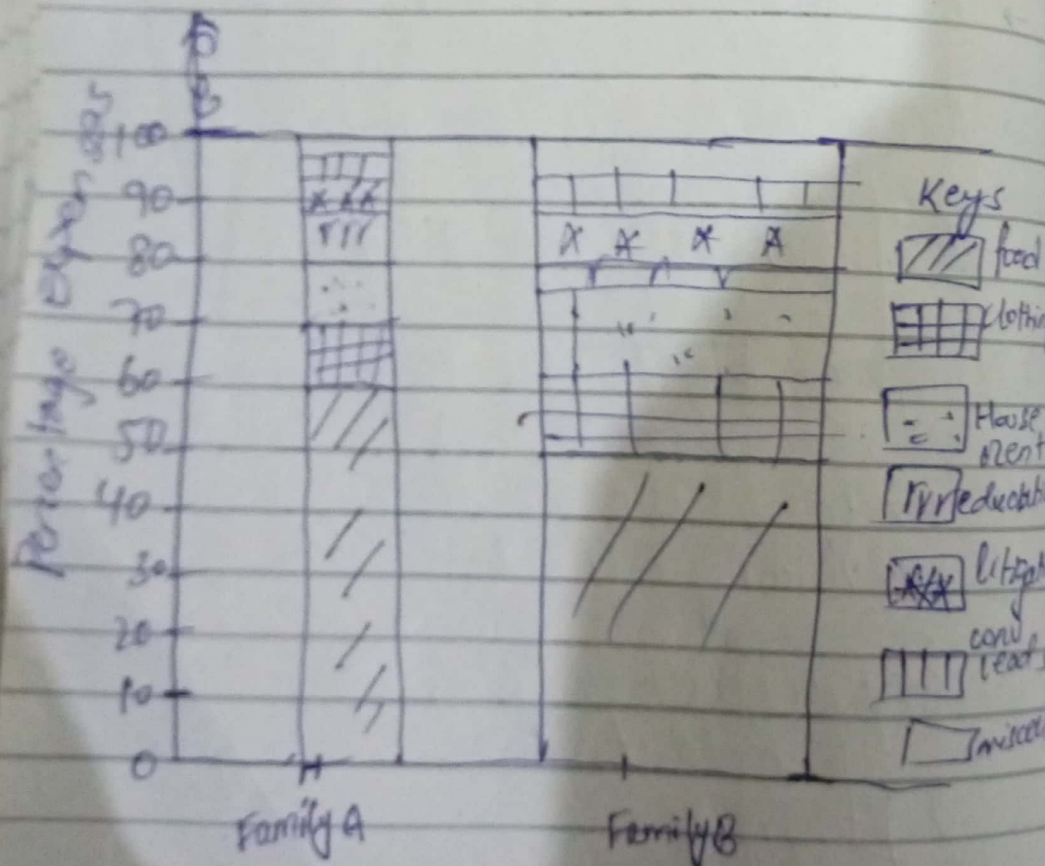


23-10-24 STAT Lec 05 Wednesday
 * Rectangles & Sub-divided Rectangles

Percentage Bar Diagram:

Items	Family A	Family B
Food	16.5% 60% 2	60 50% 50%
Clothing	10% 7% 4	4 4.7 61.7
House Rent	10% 8% 4	16 13.3 75+
Education	7.5% 8% 3	6 5 80+
Litigation	5% 12% 2	10 8.3 88.3+
Conventional needs	2.5% 1% 1	6 5 93.3+
Miscellaneous	5% 10% 2	8 6.7 100+
	100% 40	120 100

Percentage Bar Diagram:



family A family B
 total 120
 40 :
 1 : 3

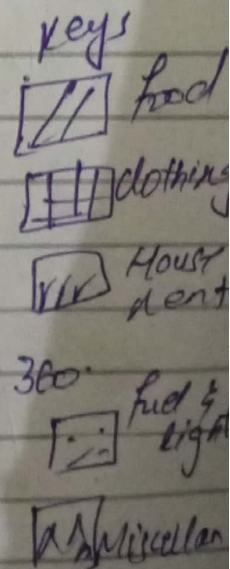
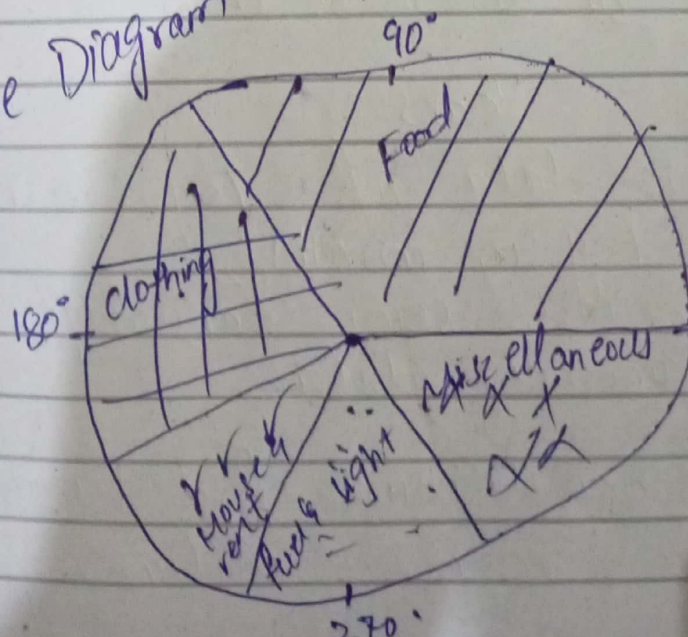
if family A takes 10 box then B takes 30 boxes.

① Pie - Diagram:

Angle of a sector = $\frac{\text{Component Part}}{\text{whole quantity}} \times 360^\circ$
 Example

Items	Exp.	Angle of a sector	Cumulative angles
Food	50	$\frac{50}{150} \times 360^\circ = 120^\circ$	120°
Clothing	30	= 72°	192°
House rent	20	= 48°	240°
Fuel & light	15	= 36°	276°
Miscellaneous	35	= 84°	360°
Total	150	360°	

Pie Diagram

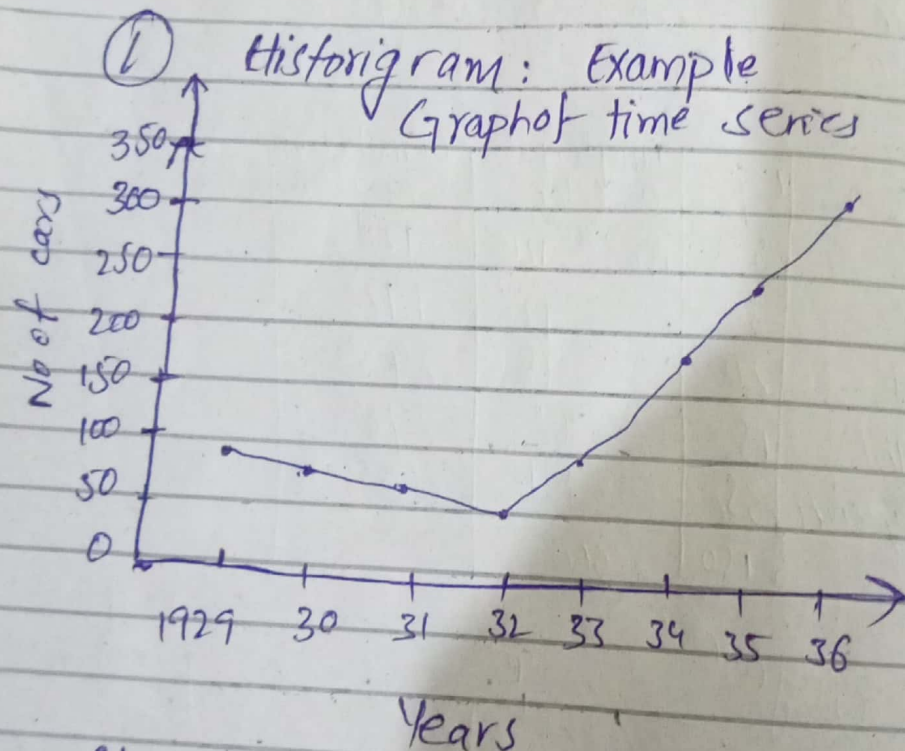


④ Graphs:

① Histogram or
graph of time series

② Graph of frequency
Distribution:

- i- Histogram
- ii- frequency curve
- iii- frequency polygon
- iv- cumulative frequency curve or
OGIVE



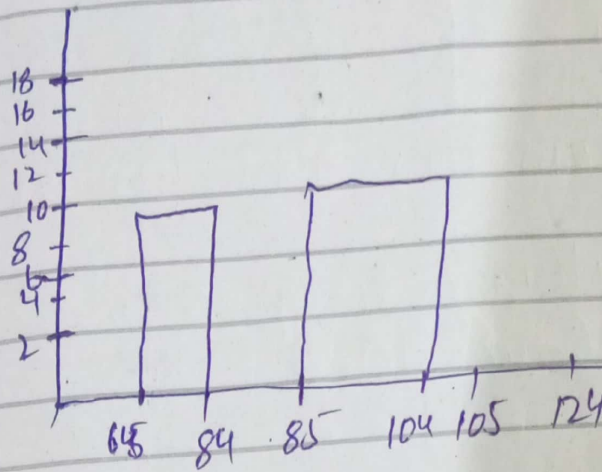
it is data of regular interval
of time (year, monthly),
also called semi-annually or
weekly annually data.

(time series is name of data with
regular interval of time).

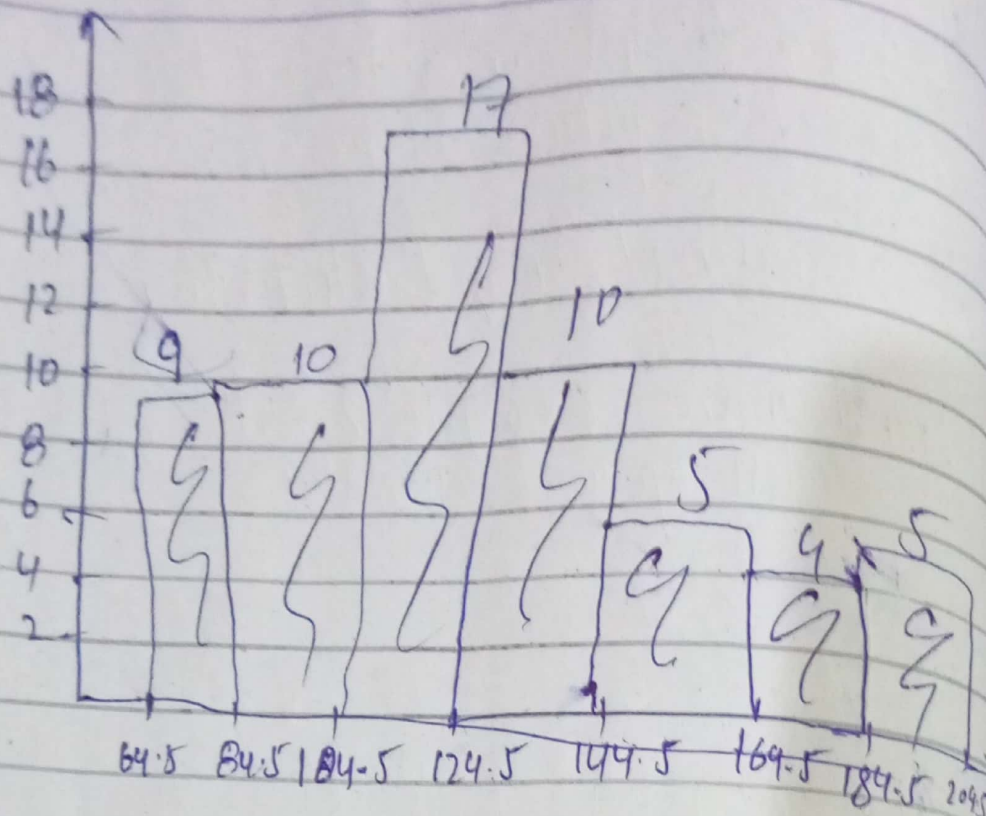
you should mention years if not mentioned in an interval b/c it is data of regular intervals.

(2) Graph of freq Distribution:

(1) Histogram: Example: 2.3: (Use of class boundaries)
Histogram of gp data,



you will get data in form of gp data.



It is making on gp data & on class boundaries.

(ii) freq. curve (joining the corner points)

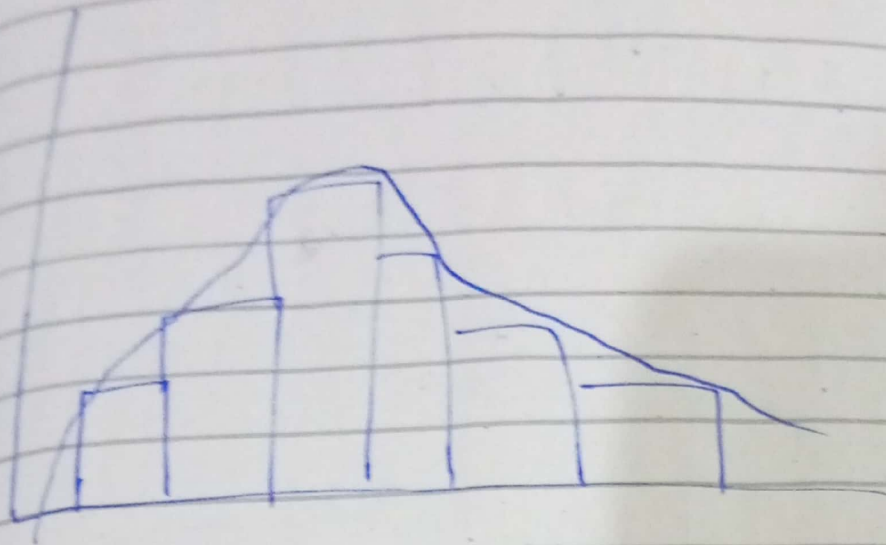
25-10-2024

Shubham

Friday

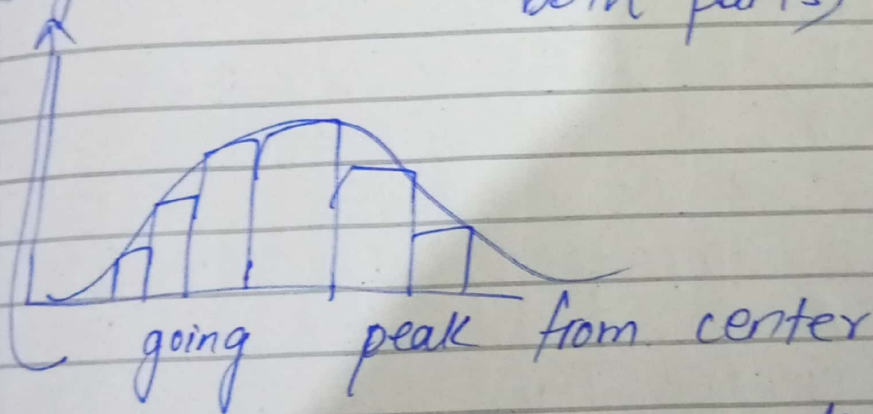
Thursday

→ (ii) frequency curve: (joining corner points)

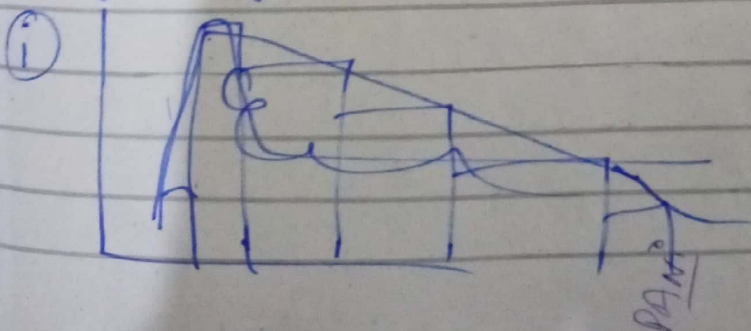


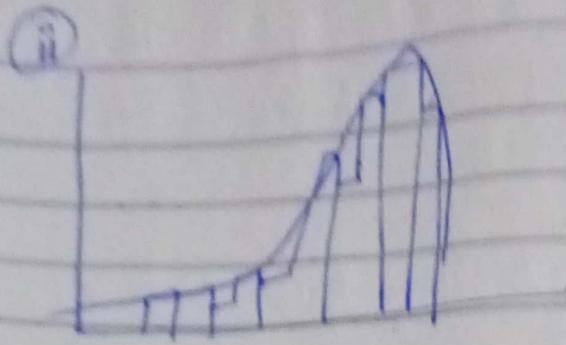
(joining the corner points)

it is called symmetrical
→ Symmetrical (divides data equally from both parts)

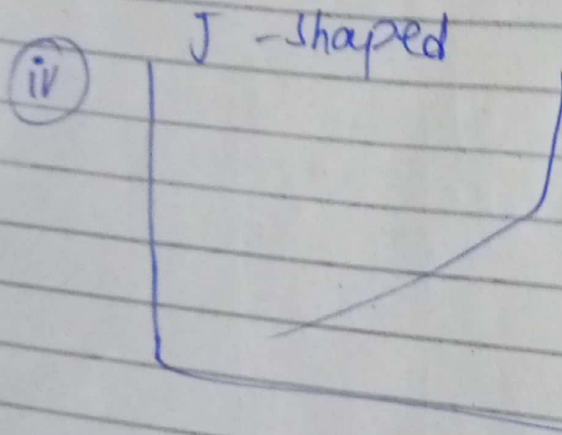
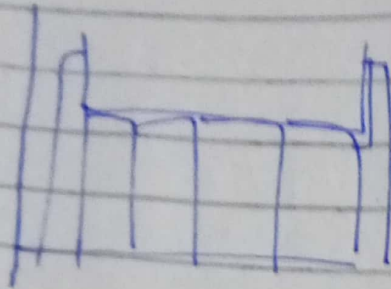


→ Asymmetrical (not symmetrical)
i) positively skewed (beh jana)
ii) negatively skewed:





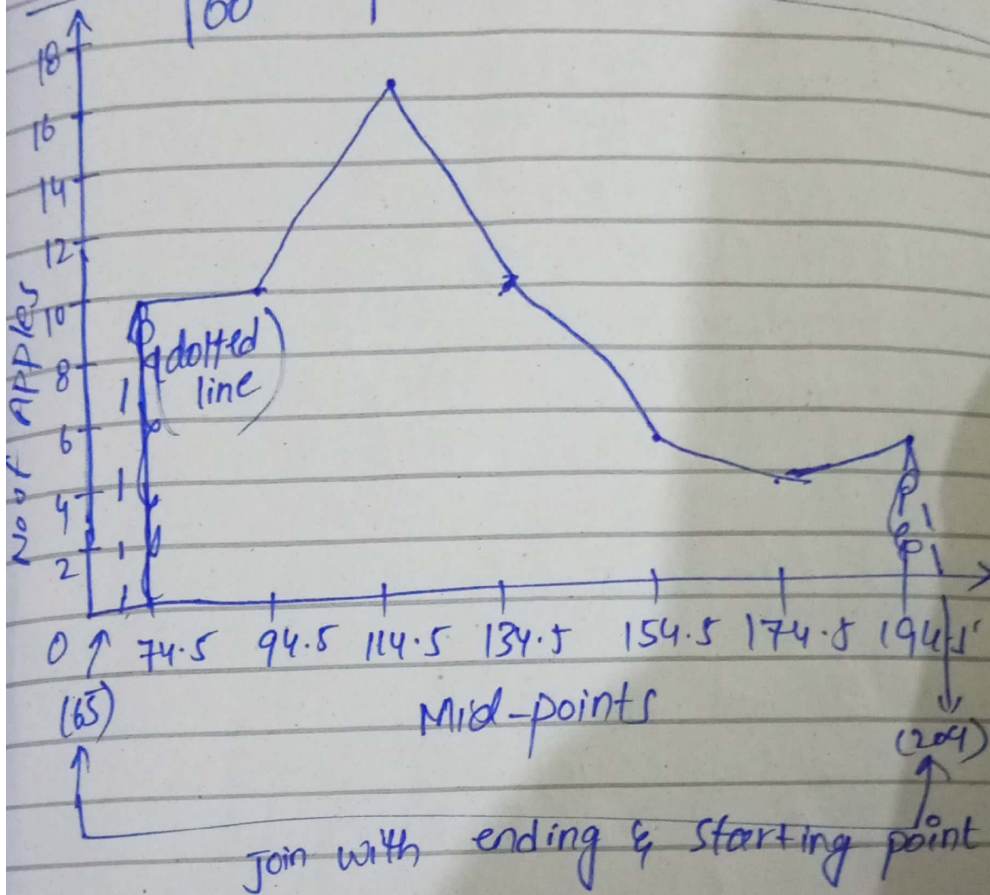
(iii) U shaped:



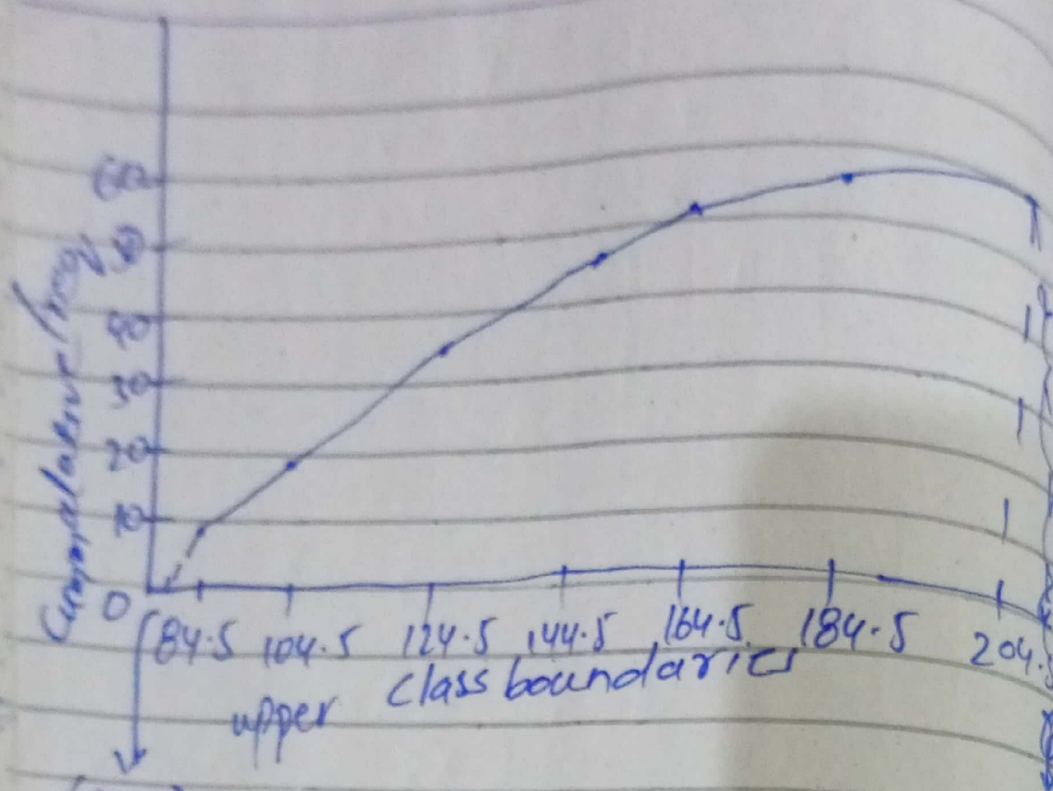
most important are (i), (ii) & (iii)

(iii) frequency polygon:
(use of midpoints):

Weight	No. of Apples (freq)	Midpoints	C.B	C.F
65-84	9	74.5	64.5-84.5	9
85-104	10	94.5	84.5-104.5	19
105-124	17	114.5	104.5-124.5	36
125-144	10	134.5	124.5-144.5	46
145-164	5	154.5	144.5-164.5	51
165-184	4	174.5	164.5-184.5	55
185-204	5	194.5	184.5-204.5	60
	60			



② Cumulative freq polygon (OGIVE)
 (use of upper class boundaries)
 use C.B



(64.5) ↗ closed boundary with lower class limit

chapter no 03:

30-10-2024

STAT lec no 66

Wednesday

① Measure of central tendency:

Measure of Averages:

Measure of locations:

or
Measure of Positions:

• Types of Averages:

- i) Arithmetic Mean (A.M):
- ii) Geometric Mean (G.M):
- iii) Harmonic Mean (H.M):
- iv) Median
- v) Mode

i) Arithmetic Mean (A.M):

• Symbolically:

$\bar{X} \rightarrow$ sample mean

$\mu \rightarrow$ population mean

Methods	for ungroup data	for group data
Direct Method	$\bar{x} = \frac{\sum x}{n}$	$\bar{x} = \frac{\sum fx}{\sum f}$
Indirect or Shortcut Method	$\bar{x} = A + \frac{\sum d}{n}$	$\bar{x} = A + \frac{\sum fd}{\sum f}$
Step Deviation or coding method	X	$\bar{x} = A + \left[\frac{\sum fu}{\sum f} \right] \times h$

for $\bar{x} = \frac{\sum x}{n}$

population mean:

(i) $\mu = \frac{\sum x}{N}$

(ii) $\mu = A + \frac{\sum d}{N}$

Explanation:

- $\sum \rightarrow$ summation
- $n \rightarrow$ sample size
- $x \rightarrow$ ~~sample~~ for ungroup data (values)
for group data (mid-points)

- $f \rightarrow$ frequency
- $A \rightarrow$ Assumed value of " x "
- $D \rightarrow x - A$
 $D =$ Deviation of values from arbitrary origin

$$U \rightarrow \frac{x-A}{h} = \frac{D}{h}$$

$h \rightarrow$ class width/height

① Example for ungrouped Data:
09, 19, 11, 15, 08, 12, 13, 16, 0, 10

Find A.M = ?
solution

$\rightarrow$ By Direct Method:

$$\bar{x} = \frac{\sum x}{n} = \frac{9+19+11+15+8+12+13+16+0+10}{10}$$

$$\boxed{\bar{x} = 11.3}$$

$\rightarrow$ By Short cut Method:

$$\bar{x} = A + \frac{\sum D}{n} = A + \frac{\sum (x-A)}{n}$$

$\therefore$ If you have more than things in summation, draw table

X	$D = (X - A) = (X - 8)$
9	1
11	11
11	3
15	7
8	0
12	4
13	5
16	8
0	-8
10	2
	33

$$\bar{X} = A + \frac{\sum D}{n}$$

$$= 8 + \frac{33}{10}$$

$$\bar{X} = 11.3$$

⑪ Example for Grouped Data
~~marks~~ find A.M = ?

Marks	No. of students	Mid points	fx
3-6	3	4.5	13.5
7-10	12	8.5	102
11-14	16	12.5	200
15-18	13	16.5	214.5
19-22	10	20.5	205
23-26	7	24.5	171.5
	61		906.5

→ By Direct Method

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{906.5}{61} = 14.86$$

→ By shortcut Method:

$$\bar{x} = A + \frac{\sum fd}{\sum f}$$

Marks	No. of student (f)	$D = x - 12.5$	$D = (x - A) = (x - 12.5)$	fd	$\frac{fd}{h}$
3-6	3	4.5	-8	-24	-2
7-10	12	8.5	-4	-48	-1
11-14	16	12.5	0	0	0
15-18	13	16.5	4	52	1
19-22	10	20.5	8	80	2
23-26	7	24.5	12	84	3
	61			144	

$$\bar{x} = 12.5 + \frac{144}{61} = 12.5 + 2.36$$

$$\boxed{\bar{x} = 14.86}$$

(iii) By step Deviation

$$\bar{x} = A + \left(\frac{\sum fu}{\sum f} \right) h$$

$$= 12.5 + \left(\frac{36}{61} \right) \times 4$$

$$\boxed{\bar{x} = 14.86}$$

fu
-6
-12
0
13
20
21
36

(*) Properties of A.M:

(1) Sum of Deviation from Mean ($\bar{x}$) always equal to zero

(i) $\sum (x - \bar{x}) = 0$ (for ungp data)

(ii) $\sum f(x - \bar{x}) = 0$ (for gp data)

prove it

x	$(x - \bar{x})$	$(x - \bar{x})^2$	$(x - A)$	x_n
$\left. \begin{array}{c} \{ \\ \{ \\ \{ \end{array} \right\}$	$\left. \begin{array}{c} \{ \\ \{ \\ \{ \end{array} \right\}$	$\left. \begin{array}{c} \{ \\ \{ \\ \{ \end{array} \right\}$	$\left. \begin{array}{c} \{ \\ \{ \\ \{ \end{array} \right\}$	$\left. \begin{array}{c} \{ \\ \{ \\ \{ \end{array} \right\}$
	$\sum (x - \bar{x}) = 0$			

(2) Sum of squared deviation from mean ($\bar{x}$) is minimum.

(i) $\sum (x - \bar{x})^2 \leq \sum (x - A)^2$ ^{D^2}
(ungp data)

(ii) $\sum f(x - \bar{x})^2 \leq \sum f(x - A)^2 \rightarrow fD^2$
(gp data)

(3) if $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_k$ are k arithmetic means with $n_1, n_2, n_3, \dots, n_k$ sample sizes, then their combined mean ($\bar{x}_c$) will be find

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3 + \dots + n_k \bar{x}_k}{n_1 + n_2 + n_3 + \dots + n_k}$$

Subject	Sample mean ($\bar{x}$)	Sample size (n)
P&S	$\bar{x}_1 = 63$	$n_1 = 20$
ADA	$\bar{x}_2 = 56$	$n_2 = 40$

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

→ Grouped Data

Marks	f	x	fx	$(x - \bar{x})$	$\sqrt{x - \bar{x}}$	$f(x - \bar{x})$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
3-6	3	4.5	13.5	-10.36	-3.21	-31.08	107.33	307
7-10	12	8.5	102	-6.36	-2.52	-76.32	16.02	192
11-14	18	12.5	225	-2.36	-1.53	-42.48	0.58	10
15-18	13	16.5	214.5	1.64	1.28	21.32	2.10	27
19-22	10	20.5	205	5.64	2.37	56.4	13.35	133
23-26	7	24.5	171.5	9.64	3.10	67.48	30.85	216
	61		906.5			0.04	16405.58	190

$$\sum f(x - \bar{x}) = 0$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{906.5}{61} = 14.86$$

$$\begin{aligned} \sum f(x - \bar{x})^2 &= 16405.58 \\ \sum f(x - A)^2 &= 19040 \\ \sum f(x - \bar{x})^2 &\leq \sum f(x - A)^2 \end{aligned}$$

→ Ungrouped Data:

x	$(x - \bar{x})$	$(x - \bar{x})^2$	$(x - A=12)$	$(x - A)^2$
9	-2.3	5.29	-3.8	14.44
19	7.7	59.29	7	49
11	-0.3	0.09	-1.8	3.24
15	3.7	13.69	3	9
8	-3.3	10.89	-4.8	23.04
12	0.7	0.49	0	0
13	1.7	2.89	1	1
16	4.7	22.09	4	16
0	-11.3	127.69	-12.8	163.84
10	-1.3	1.69	-2.8	7.84
	0	244.47		258.54
$\bar{x} = \frac{\sum x}{n}$				

$$\sum (x - \bar{x}) = 0$$

6393

$$\sum (x - \bar{x})^2 = 244.1$$

$$\sum (x - A)^2 = \cancel{250.5} 249$$

So

$$\sum (x - \bar{x})^2 \leq \sum (x - A)^2 \text{ proved}$$

1-11-2024 STAT 101 00 07 Friday

(ii) Geometric Mean (G.M.):
 It is the root of product of observation of items / values

(a) for ungrouped Data:-

$$\rightarrow G.M = (x_1 \times x_2 \times x_3 \dots \times x_n)^{1/n}$$

$$\text{OR}$$

$$\rightarrow G.M = \text{antilog} \left[\frac{\sum \log x}{n} \right]$$

(b) for Grouped Data:-

$$\rightarrow G.M = (x_1^{f_1} \times x_2^{f_2} \times x_3^{f_3} \dots \times x_n^{f_n})^{1/\sum f}$$

$$\text{OR}$$

$$\rightarrow G.M = \text{antilog} \left[\frac{\sum f \log x}{\sum f} \right]$$

(iii) Harmonic Mean (H.M.):

(a) for ungroup data:-

$$\rightarrow H.M = \frac{n}{\sum \left(\frac{1}{x} \right)}$$

for group data

$$\rightarrow H.M = \frac{\sum f}{\sum \left(\frac{f}{x} \right)}$$

for avg speed, we do harmonic mean

from previous example of ungroup data

X	log X	$\left(\frac{1}{X} \right)$
9	0.9542	0.1111
19	1.2788	0.0526
11	1.0414	0.0909
15	1.1761	0.0667
8	0.9031	0.1250
12	1.0792	0.0833
13	1.1139	0.0769
16	1.2041	0.0625
0	1	∞
10	1	0.100
	10.7508	$= \infty$

find G.M & H.M

$$\rightarrow G.M = \text{antilog} \left[\frac{\sum f \log x}{\sum f} \right]$$

$$= \text{antilog} \left[\frac{10.7508}{10} \right]$$

OR

$$G.M = (9 \times 19 \times 11 \times 15 \times 8 \times 12 \times 13 \times 16)^{1/8}$$

$$G.M = \textcircled{0}$$

'0' does not in G.M

$\rightarrow H.M$:

$$H.M = \frac{n}{\sum \left(\frac{1}{x} \right)}$$

its answer will be infinity.

"speed can't be zero"
 → from previous example of b.p data

Marital	f	x	log x	f log x	$\frac{f}{N}$
3-6	8	4.5	0.6532	5.2256	0.6667
7-10	12	8.5	0.9294	11.1528	1.4118
11-14	16	12.5	1.0969	17.5504	1.28
15-18	13	16.5	1.2175	15.8275	0.7879
19-22	10	20.5	1.3106	13.106	0.4878
23-26	7	24.5	1.3899	9.7293	0.2857
	61			69.3947	4.9199

$$\rightarrow G.M = \text{antilog} \left[\frac{\sum f \log x}{\sum f} \right] = \text{antilog} \left[\frac{69.3947}{61} \right]$$

$$G.M = 13.7283$$

$$\rightarrow H.M = \frac{\sum f}{\sum \left(\frac{f}{x} \right)} = \frac{61}{4.9199} = 12.3986$$

ii) Median ($\tilde{x}$) :-
 "middle value of the data, after arranging the data in ascending order".

→ for ungroup data :-

when number of values are odd
 (i.e., $n = \text{odd}$)

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{value in "arranged data"}$$

when number of values are even
 (i.e., $n = \text{even}$)

$$\text{Median} = \frac{1}{2} \left[\left(\frac{n}{2} \right)^{\text{th}} \text{value in arranged data} + \left(\frac{n+1}{2} \right)^{\text{th}} \text{value in arranged data} \right]$$

Note: "it divides the data into two equal parts".

iii) Quartiles :-
 "it divides the data into four equal parts".

Q_1 = lower quartile
 or
 first quartile

$$Q_1 = 1 \times \left(\frac{n+1}{4} \right)^{\text{th}} \text{value in arranged data}$$

$$Q_2 = \left(\frac{n+1}{4} \right)^{\text{th}} = \text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{value in arranged order}$$

Q_3 = upper quartile
 or
 third quartile

$$Q_3 = 3 \times \left(\frac{n+1}{4} \right)^{\text{th}} \text{value in arranged data}$$

"why not Q_4 ?"

③ Deciles:
"it divides the data into ten equal parts"

$$D_1 = \left(\frac{n+1}{10} \right)^{\text{th}} \text{ value in arranged data}$$

$$D_2 = \left(\frac{2(n+1)}{10} \right)^{\text{th}} \text{ value in arranged data}$$

$$D_5 = 5 \left(\frac{n+1}{10} \right)^{\text{th}} \Rightarrow \text{Median}$$

$$D_9 = 9 \left(\frac{n+1}{10} \right)^{\text{th}} \dots$$

④ Percentiles:
"it divides the data into 100 equal parts"

$$P_1 = \left(\frac{n+1}{100} \right)^{\text{th}} \dots$$

$$P_2 = 2 \left(\frac{n+1}{100} \right)^{\text{th}} \dots$$

$$P_{50} = 50 \left(\frac{n+1}{100} \right)^{\text{th}} \Rightarrow \text{Median}$$

$$P_{87} = 87 \left(\frac{n+1}{100} \right)^{\text{th}} \dots$$

$$P_{99} = 99 \left(\frac{n+1}{100} \right)^{\text{th}} \dots$$

⑤ Example for ungrouped Data:-

9, 19, 11, 15, 8, 12, 13, 16, 0, 10.

Arranged Data:

0, 8, 9, 10, 11, 12, 13, 15, 16, 19

$$\text{Median } (\bar{x}) = \frac{1}{2} \left[\left(\frac{n}{2} \right)^{\text{th}} \text{ value} + \left(\frac{n+1}{2} \right)^{\text{th}} \text{ value} \right]$$

$$= \frac{1}{2} \left[\frac{10^{\text{th}}}{2} + \left(\frac{11}{2} \right)^{\text{th}} \right]$$

$$= \frac{1}{2} \left(\frac{10}{2} + 5^{\text{th}} + 6^{\text{th}} \right)$$

$$= \frac{1}{2} (11 + 12)$$

$$\boxed{\bar{x} = 11.5}$$

Quartile:

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{\text{th}} \text{ value} = \frac{3}{4} \left(\frac{10+1}{4} \right) = 3 \left(\frac{11}{4} \right)^{\text{th}}$$

$$= 3(2.75)^{\text{th}} \text{ value}$$

$$Q_3 = (8.25)^{\text{th}} \text{ value}$$

$$Q_3 = 8^{\text{th}} \text{ value} + 0.25(9^{\text{th}} \text{ value} - 8^{\text{th}} \text{ value})$$

$$= 15 + 0.25(16 - 15)$$

$$= 15 + 0.25$$

$$Q_3 = 15.25$$

"for checking correct ans will be in its domain".

$$P_{87} = 87 \left(\frac{n+1}{100} \right)^{\text{th}} \text{ value}$$

$$= 87 \left(\frac{10+1}{100} \right)^{\text{th}} \text{ value}$$

$$= 87 (0.11)^{\text{th}} \text{ value}$$

$$= (9.57)^{\text{th}} \text{ value}$$

$$= (9^{\text{th}} \text{ value} + 0.57(10^{\text{th}} - 9^{\text{th}}))$$

$$= 16 + 0.57(19 - 16)$$

$$= 16 + 1.71$$

$$P_{87} = 17.71$$

for

$$Q_4 = 4 \left(\frac{n+1}{4} \right)^{\text{th}} \text{ value}$$

= 11th value, so not possible

(*) for grouped Data:-

$$\rightarrow \text{Median} = L + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$\rightarrow Q_1 = L + \frac{h}{f} \left(\frac{n}{4} - c \right)$$

$$\rightarrow Q_2 = L + \frac{h}{f} \left(\frac{2n}{4} - c \right) \equiv \text{Median}$$

$$\rightarrow Q_3 = L + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

$$\rightarrow D_1 = L + \frac{h}{f} \left(\frac{n}{10} - c \right)$$

!

$$\rightarrow D_3 = L + \frac{h}{f} \left(\frac{3n}{10} - c \right)$$

$$\rightarrow D_5 = \text{Median}$$

$$\rightarrow D_9 = L + \frac{h}{f} \left(\frac{9n}{10} - c \right)$$

$$\rightarrow P_1 = L + \frac{h}{f} \left(\frac{n}{100} - c \right)$$

$$P_{33} = L + \frac{h}{f} \left(\frac{33n}{100} - c \right)$$

$P_{50} = \text{Median}$

$$P_{99} = L + \frac{h}{f} \left(\frac{99n}{100} - c \right)$$

Question for Grouped Data:

Marks	f	C.B	Cumulative freq
3-6	3	2.5-6.5	3
7-10	12	6.5-10.5	15 = c
11-14	16	10.5-14.5	31
15-18	13	14.5-18.5	44
19-22	10	18.5-22.5	54
23-26	7	22.5-26.5	61 = n
	61		

for median, we need C.B

$$\text{Median} = L + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$\frac{n}{2} = \frac{61}{2} = 30.5 \text{ lies in "10.5-14.5"}$$

$$L = \text{lower class bound} = 10.5$$

$$h = \text{class width} = 4$$

$$f = \text{freq} = 16$$

$$c = \text{previous c.f} = 15$$

$$= 10.5 + \frac{4}{16} \left(\frac{30.5 - 15}{1} \right)$$

$$\bar{x} = 14.375$$

Same for all quartiles, percentiles & deciles

$$Q_3 = ?$$

$$= L + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

$$= 10.5 + \frac{4}{16} \left(\frac{3 \times 61}{4} - 15 \right)$$

$$\text{lies in } 18.5 - 22.5$$

$$L = 18.5, h = 4, f = 10, c = 44$$

→ find $D_7, D_8 = ?$ $P_{87}, ?$

8-11-2024

STAT Lec no 08

Friday

5) Mode: ($\hat{x}$) :-

"most frequent or most common or most repeated or most occurring value in a data set."

• for ungp data:

→ ex: 01 →

3, 5, 10, 12, 3, 7

Mode ($\hat{x}$) = 3 (unimode)

→ ex: 02 →

5, 4, 9, 10, 5, 8, 4, 1

Mode ($\hat{x}$) = 5, 4 (Bimode)

→ ex: 03 →

10, 14, 6, 8, 7, 3, 10

8, 9, 12, 14, 6

Mode ($\hat{x}$) = 6, 10, 8, 14 (Multimode)

→ ex: 04 →

10, 9, 6, 3, 12, 14, 1, 8

Mode = no mode

→ ex: 05 →

13, 8, 10, 6, 4, 10, 12,

9, 13, 4, 5, 10, 15, 3,

12, 8, 6, 5, 10, 19, 8,

17, 10, 3, 7, 16.

Mode ($\hat{x}$) = 10

for gpd data:

Marks	f	C.B
3-6	3	2.5-6.5
7-10	18	6.5-10.5
11-14	16	10.5-14.5
15-18	13	14.5-18.5
19-22	10	18.5-22.5
23-26	7	22.5-26.5
	61	

→ modal class

sol:

$$\text{Mode } (L) = L + \left[\frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \right] \times h$$

$$f_m = 16, f_1 = 12, f_2 = 13, h = 4, L = 10.5$$

$$= 10.5 + \left[\frac{16 - 12}{(16 - 12) + (16 - 13)} \right] \times 4$$

$$= 10.5 + \left[\frac{4}{4 + 03} \right] \times 4$$

$$10.5 + \left[\frac{4}{07} \right] \times 4$$

$$10.5 + \frac{16}{07}$$

$$10.5 + 2.2857$$

$$12.7857$$

$$\hat{x} = 12.7857$$

"this value will be in modal class."

13-11-24

Chapter no 04 Stat lec no 09 Wedne
Chapter no 04:
"Measure of Dispersion" = "Sto."

~~3, 5, 10, 12, 3, 7~~

3, 5, 10, 12, 3, 7

$$\hat{x} = 3$$

"values have variation"
measure of avg = $\hat{x} = 3$

has value 10 will done for fastly
they are of two types
"the value"

• Absolute Measure of Deviations:

1- Range (R)
2- Quartile Deviation (QD) or
Semi-inter quartile range

3- Mean Deviation (M.D) OR
Average Deviation or
Absolute Deviation

4- Standard Deviation (S.D)
AND
Variance

calculate percentage

• Relative Measure of Dispersion:

- 1) Coefficient of Range
- 2) Coefficient of Quartile Deviation
- 3) Coefficient of Mean Deviation
- 4) Coefficient of Variation (C.V)

① Range (R):

$$R = X_m - X_o$$

• from previous example of ungroup data
 $X_m = 12, X_o = 3$

$$\Rightarrow R = 12 - 3$$
$$[R = 9]$$

$$\text{Coefficient of Range} = \frac{X_m - X_o}{X_m + X_o} = \frac{12 - 3}{12 + 3}$$
$$= \frac{9}{15} = \left(\frac{3}{5} \right)$$

"Absolute always come range b/w 0-1"

• from previous example of gp data:
 $X = \text{mid-point, High M.D, Low M.D}$
 $X_m = 24.5, X_o = 4.5$

in book ($X_m = \text{highest class boundary} = 26.5$)
($X_o = \text{lowest class boundary} = 2.5$)

$$R = 24.5 - 4.5 = 20$$

$$\text{Coeff of Range} = \frac{x_m - x_o}{x_m + x_o} = \frac{24.5 - 4.5}{24.5 + 4.5} = \frac{20}{29} = 0.6897$$

② Quartile Deviation (Q.D):

$$Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$\text{Coeff of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

"Same formulae for ungp & gp data"

• for ungp data:

$$Q_1 = \left(\frac{n+1}{4} \right)^{\text{th value}}$$

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{\text{th value}}$$

• for gp data:

$$Q_1 = l + \frac{h}{f} \left(\frac{n}{4} - c \right)$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

③ Mean Deviation (M.D):
Absolute Dispersion

Methods	for ungp data	for gp data
M.D from $(\bar{x})$ Mean	$\frac{\sum x - \bar{x} }{n}$	$\frac{\sum f x - \bar{x} }{\sum f}$
M.D from Median $(\tilde{x})$	$\frac{\sum x - \tilde{x} }{n}$	$\frac{\sum f x - \tilde{x} }{\sum f}$
M.D from Mode $(\hat{x})$	$\frac{\sum x - \hat{x} }{n}$	$\frac{\sum f x - \hat{x} }{\sum f}$

Deviation: $(x - \text{constant})$

• Coefficients:

$$\rightarrow \text{coefficient of M.D by using Mean}(\bar{x}) = \frac{\text{M.D from Mean}(\bar{x})}{\text{Mean}(\bar{x})}$$

$$\rightarrow \text{coefficient of M.D by using Median}(\tilde{x}) = \frac{\text{M.D from Median}(\tilde{x})}{\text{Median}(\tilde{x})}$$

$$\rightarrow \text{coefficient of M.D by using Mode}(\hat{x}) = \frac{\text{M.D from Mode}(\hat{x})}{\text{Mode}(\hat{x})}$$

⑤ from previous example of wggp data:

3, 5, 10, 12, 3, 7

find M.D from Median ($\bar{X}$) &

its coefficient?

sol:

sol: As M-D from Median $(\bar{x}) = \frac{\sum (x - \bar{x})}{n}$

$$\bar{X} = \frac{1}{2} \left[\left(\frac{n}{2} \right)^{\text{th value}} + \left(\frac{n+2}{2} \right)^{\text{th value}} \right]$$

$$= \frac{1}{2} [(3)^{\text{th}} \text{ value} + 4^{\text{th}} \text{ value}]$$

Arrange

3, 5, 7, 10, 12, 13

$$= \frac{1}{2} (8 + 10)$$

$$= \frac{1}{2} \ln 18$$

$$x = 886$$

x	$x - \bar{x}$
3	3
5	1
7	1
10	4
12	6
13	7 + 3
15	8

$$M.D = \frac{\sum |x - \bar{x}|}{n} = \frac{18}{6} = 3$$

$$\text{Coeff of } \frac{M.D}{M.D} = \frac{M.D}{\cancel{M.D}} = \frac{3}{6} = \frac{1}{2} = 0.5$$

(*) for Gp Data:

[illegible]

M.D from Mode $(\hat{x}) = \frac{\sum f|x - \hat{x}|}{\sum f}$

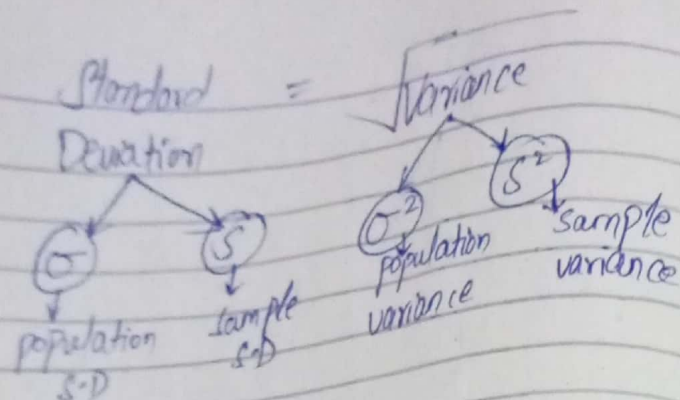
$$\hat{x} = 1 + \left[\frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \right] \times h = 12.78$$

↓
from previous

from previous
qp data

$$\text{Coeff} = \frac{M \cdot D}{x}$$

⑨ Standard Deviation (S.D) &
Variance:



$$S = \sqrt{S^2}$$

$S = S$ so relation proved

Methods	for ungp data	for gp data
Direct Method - I	$S^2 = \frac{\sum (x - \bar{x})^2}{n}$ $S = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$	$S^2 = \frac{\sum f(x - \bar{x})^2}{\sum f}$
Direct Method - II	$S^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$ $\bar{x}$	$S^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2$
Indirect Method or Shortcut Method	$S^2 = \frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2$ $D = x - A$	$S^2 = \frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2$
Step Deviation Method or Coding Method	$S^2 = \left[\frac{\sum fu^2}{\sum f} - \left(\frac{\sum fu}{\sum f}\right)^2 \right] \times h^2$ $u = \frac{x - A}{h} = \frac{D}{h}$	

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad (\mu = \text{pop mean})$$

(N = pop)

Population Variance $\left\{ \begin{array}{l} S.D \& var \text{ can never be } -ve \end{array} \right\}$

Do S.D & var for ungp & gp data.

Relative Dispersion:

Coeff of variation (C.V) = $\left[\frac{S}{\bar{x}} \times 100 \right]$

= $\%$

5-11-2024

Stat lec no 10

Friday

Moments:

Sample moments (m_r)

Population moments (μ_r)

→ Moments about mean (m_r)

→ Moments about arbitrary origin (say "A") (m'_r)

Methods	For ungrouped data	for sample for grouped data
Moments about mean ($\bar{x}$)	$m_r = \frac{\sum (x - \bar{x})^r}{n}$	$m_r = \frac{\sum f(x - \bar{x})^r}{\sum f}$
Moments about arbitrary origin (say "A")	$m'_r = \frac{\sum (x - A)^r}{n}$ or $m'_r = \frac{\sum D^r}{n}$ $\therefore D = (x - A)$	$m'_r = \frac{\sum f(x - A)^r}{\sum f}$ or $m'_r = \frac{\sum fD^r}{\sum f}$ $\therefore D = (x - A)$

• for example: in case of population

$$\mu_r = \frac{\sum (x - \mu)^r}{N}$$

$r = 0, 1, 2, 3, 4$ (4th order moment)

→ find zero order moment

put $r=0$

$$m_0 = \frac{\sum (x - \bar{x})^0}{n} = \frac{\sum (1)}{n} = \frac{n}{n} = 1$$

→ 1st order moment

put $r=1$

$$m_1 = \frac{\sum (x - \bar{x})^1}{n} = \frac{0}{n} = 0$$

$$\therefore \sum (x - \bar{x}) = 0$$

→ 2nd order moment

put $r=2$

$$m_2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{5^2}{5} = 5$$

"that means variance & 2nd moment are same"

$$m_3 = \frac{\sum (x - \bar{x})^3}{n} \quad \left\{ \begin{array}{l} \text{used for} \\ \text{distribution of} \\ \text{shapes (i.e.,} \\ \text{leptokurtic, etc)} \end{array} \right.$$

$$m_4 = \frac{\sum (x - \bar{x})^4}{n}$$

for example letn ungp data):
9, 4, 6, 8, 13

find 3rd order moment about mean ($\bar{x}$) & 2nd order moment about $x=A$ & $A=4$

$$m_r = \frac{\sum (x - \bar{x})^r}{n}$$

if A is changed, ans will be change

x	$(x - \bar{x})$	$(x - \bar{x})^2$	$(x - A)^1$	$(x - A)^2$
9	1	1	5	25
4	-4	-64	0	0
6	-2	-8	2	4
8	0	0	4	16
13	5	125	9	81
40	0	54		126

$$\bar{x} = \frac{\sum x}{n} = \frac{40}{5} = 8$$

$$m_3 = \frac{\sum (x - \bar{x})^3}{n} = \frac{54}{5} = 10.8$$

$$m_2' = \frac{\sum (x - A)^2}{n} = \frac{126}{5} = 25.2$$

[illegible]

$$m'_3 = \frac{\epsilon f (x-A)^3}{\epsilon f}$$

$\Sigma f = 40, \Sigma fD = 30, \Sigma fD^2 = 100$
find m_2

* Skewness (b₁):
"to check shape"

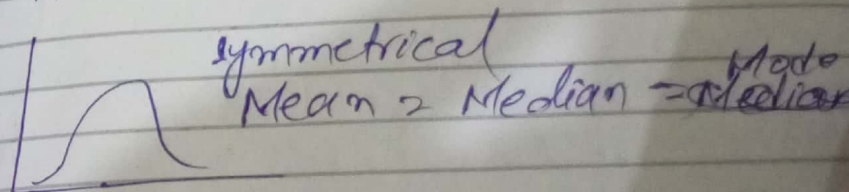
$$k = b_1 = \frac{\text{Mean-Mode}}{S}$$

- Bowley's coefficient $(-1 \text{ to } +1)$ range of skewness:

$$S_k = \frac{Q_1 + Q_3}{Q_3 - Q_1} \rightarrow \text{Median}$$

$$\therefore \text{Median} = Q_2$$

(i) for pearson
$$S_{ik} = \beta_1 = \frac{\text{Mean} - \text{Mode}}{15}$$

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$$


∴ Positively skewed:

Mean $>$ Median $>$ Mode

- Negatively skewed:

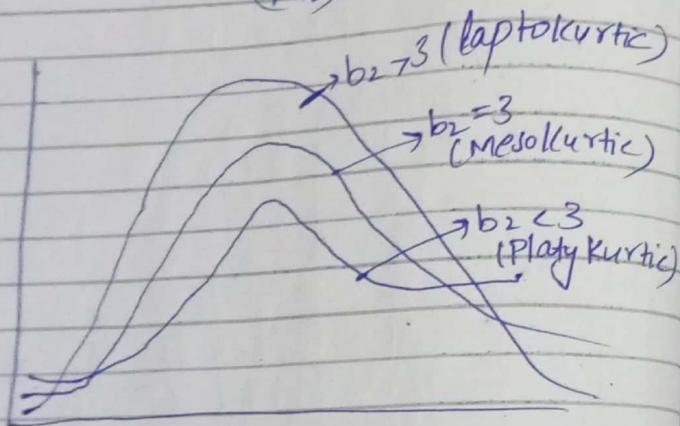
Mean < Median < Mode OR
Mode > Median > Mean

not given find coeff of skewness if distribution is

Symmetrical: $b_1 = \frac{(m_3)}{(m_2)^3}$
 Means Median = Mode

(*) Kurtosis (b_2):

$$b_2 = \frac{m_4}{(m_2)^2}$$



"Normal Distribution is mesokurtic & that is symmetrical".

20-11-24 Stat Lec no 11 Wednesday
 (*) Regression & Correlation: (chap 11)

• Regression:
 → it comes from regress.
 prediction of dependent var from independent

→ if y on x :
 the regression equation of y on x is:

$$y = a + b_{yx}x$$

(OR)

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

→ Explanation:

- y = Dependent variable
- x = independent variable
- b_{yx} = Regression coefficient (if y on x)
 OR
 b_{yx} = Slope of Regression equation

$$i) b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

OR

$$ii) b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

OR

$$iii) b_{yx} = r \frac{S_y}{S_x}$$

where $S_y = \sqrt{\frac{\sum (y - \bar{y})^2}{n}} = \sqrt{\frac{\sum y^2}{n} - \left(\frac{\sum y}{n}\right)^2}$

$$S_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

• a = Intercept of regression equation

$$\bar{y} = \frac{\sum y}{n}, \quad \bar{x} = \frac{\sum x}{n}$$

Application is predicting next 10 years marks.

→ if X on Y :
the regression equation of X on Y is:

$$X = a + b_{yx} Y$$

$$X - \bar{x} = b_{yx} (Y - \bar{y})$$

- X = Dependent variable
- Y = Independent variable

• b_{yx} = Regression coefficient (if X on Y)
= slope of regression equation

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2}$$

$$\text{or } b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

$$\text{or } b_{yx} = r \frac{S_x}{S_y}$$

a = Intercept
 $a = \bar{x} - b_{yx} \bar{y}$

⊗ Correlation:
"strength or degree of association between two variables"

Symbolically $\nearrow r$ (sample correlation)
 $\searrow \rho$ (population correlation)

$$\textcircled{i} r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

or

$$\textcircled{ii} r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

or

$$\textcircled{iii} r = \frac{\sum (x - \bar{x})(y - \bar{y})}{n S_x S_y}$$

• Properties:

⓪ It is symmetrical i.e., $r_{xy} = r_{yx}$

⓪ It lies between -1 and $+1$

⓪ It is the geometric mean (G.M) of two regression coefficients.
 $r = \sqrt{b_{yx} \times b_{xy}}$

⓪ It is independent of origin & scale
let $U = \frac{X - A}{h}$ and $V = \frac{Y - b}{K}$

then $r_{xy} = r_{UV}$

and

$$r_{uv} = \frac{n \sum uv - \sum u \sum v}{\sqrt{[n \sum u^2 - (\sum u)^2][n \sum v^2 - (\sum v)^2]}}$$

Q = following data shows the advertising cost and weekly sales

- Calculate the coefficient of correlation.
- ~~Predict~~ Make a regression line to predict weekly sales by taking Y as dependent variable.
- Show that sum of residuals is equal to zero (i.e. $\sum (y - \hat{y}) = 0$)
- Multiply each advertising ~~cost~~ ^{cost} by "2" and subtracting from "5" and each advertising ~~cost~~ ^{sales} by "3" and subtracting it from "4" then find the new correlation coefficient.

Advertising cost (X)	10	13	9	8	9
Weekly Sales (Y)	12	14	6	12	7

Solution:

X	Y	XY	X ²	Y ²	(Y - $\bar{y}$)	$\hat{y} = 0.4 + 1.031(X)$
10	12	120	100	144	1.59	10.41
13	14	182	169	196	-0.5	13.5
9	6	54	81	36	-0.38	9.38
8	12	96	64	144	-0.35	8.35
9	7	63	81	49	-0.38	9.38
49	51	515	495	569	0	

n = 5 (no of paired observations)

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

$$= \frac{5(515) - (49)(51)}{\sqrt{[5(495) - (49)^2][5(569) - (51)^2]}}$$

$$= \frac{76}{\sqrt{(74)(244)}}$$

$$= \frac{76}{134.373}$$

$$r = 0.57$$

"weekly sales are dependent"

- The regression equation to predict the weekly sales is:

$$Y = a + b_{yx} X \rightarrow (A)$$

$$b_{yx} = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

$$= \frac{76}{74} = 1.03$$

$$a = \bar{y} - b_{yx} \bar{x}$$

$$\bar{y} = \frac{\sum Y}{n} = \frac{51}{5} = 10.2$$

$$Q = 10.2 - (1.03)(9.8)$$

the estimated regression equation to predict weekly sales is:

	$U = (2x - 5)$	$V = (3y - 4)$	U	V
iii	15		22.5	
	21		441	
	.			
	.			
	.			

$$\sqrt{m^2 u^2 - (zu)^2} \left[m^2 v^2 - (zv)^2 \right]$$

$r = +0.57$
it is positive correlation & it is a moderate positive (0.0 - 0.5) weak
if (0.7 - 1) 10.5 - 0.7) moderate
-0.87 (strong -ve)
-0.47 (weak -ve)

(1) $x_0 = 61$ $x_m = 177$

$$\text{Range} = X_m - X_0 = 177 - 61 = 116$$

$$21 + 3.3 \log(100)$$

$$= 1 + 6.8 = 7.6 \approx \textcircled{8}$$

$$h = \frac{R}{m} = \frac{116}{8} = 14.5 \approx 15$$

for accumulative +

~~6~~ 0.30

$\frac{1}{0.00}$

24-12-2024 State Lec no 12

(*) Probability: can be shown in: to

"Qualitative measurement of uncertainty"
 let 'A' be any event, then probability
 'A' is denoted by $P(A)$ and is defined as
 $P(A) = \frac{\text{no. of outcomes in A}}{\text{total possible outcomes}}$

$$\Rightarrow P(A) = \frac{n(A)}{n(S)}$$

Properties:

- (i) $0 \leq P(A) \leq 1$
- (ii) $P(\emptyset) = 0$ where $\emptyset$ is null event.
- (iii) $P(S) = 1$ where 'S' is sure event.
- (iv) $P(A) = 1 - P(\bar{A}) \therefore P(A) + P(\bar{A}) = 1$
 (naah honey ki probability) $P(\bar{A}) = P(\bar{S})$

Sample space (S):
 "list of all possible outcomes"

→ Coin:

♦ if a single coin is tossed
 $S = \{H, T\}$; $n(S) = 2$

♦ if two coins are tossed
 Possible samples = (outcomes) no. of objects
 $\Rightarrow$ Possible samples = $(2)^2 = 4$

S#	C ₁	C ₂
1	H	H
2	H	T
3	T	H
4	T	T

$$\frac{4}{2} = \frac{2}{2} = 1$$

$S = \{HH, HT, TH, TT\}$; $n(S) = 4$

♦ if three coins are tossed
 possible samples = $(2)^3 = 8$

S#	C ₁	C ₂	C ₃
1	H	H	H
2	H	H	T
3	H	T	H
4	H	T	T
5	T	H	H
6	T	H	T
7	T	T	H
8	T	T	T

$$\frac{8}{2} = \frac{4}{2} = 2 = 1$$

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
 $n(S) = 8$

→ Dice:

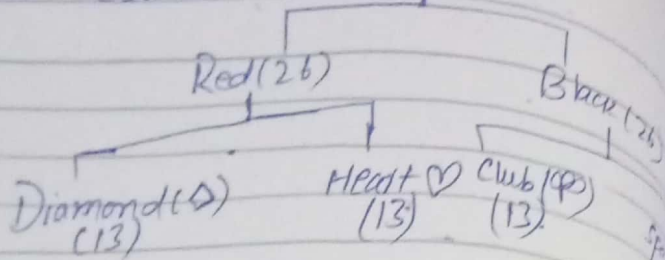
♦ if a single die is rolled
 $S = \{1, 2, 3, 4, 5, 6\}$; $n(S) = 6$

♦ if two dices are rolled
 possible samples = $(6)^2 = 36$

$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6),$
 $(2,1), (2,2), (2,3), (2,4), (2,5), (2,6),$
 $(3,1), (3,2), (3,3), (3,4), (3,5), (3,6),$
 $(4,1) \dots \dots \dots (6,6)\}$

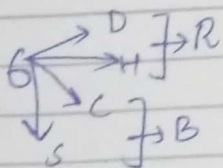
$n(S) = 36$

→ Cards: total = 52



- Number cards: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A
- picture cards: Ace (A), King (K), Queen (Q), Jack (J)
- each with four possibilities
- ↳ $(4 \times 4) = 16$

ex:



Q) if ~~two~~ ^{three} coins are tossed, what is the probability of getting:

- exactly two heads
- at least ~~one~~ ^{two} head
- at most one head
- four heads

$S = \{HHH, \dots, TTT\}$ $n(S) = 8$

i) let A be the event that denotes exactly two heads.

$A = \{HHT, HTH, TTH\}$ $n(A) = 3$

sol: $P(A) = \frac{n(A)}{n(S)} = \frac{3}{8}$

ii) let B be event that denotes at least ~~one~~ ^{two} head

$B = \{HHH, HHT, HTH, HTT, THT, TTH, TTT\}$
 $n(B) = 7$
 $P(B) = \frac{n(B)}{n(S)} = \frac{7}{8}$

iii) $P(C) = \frac{4}{8} = \frac{1}{2}$

iv) $P(D) = \frac{0}{8} = 0$
 $D = \{\}$; ϕ

Q) if two dice are rolled: what is probability of getting:

- a sum of 8
- same no on upper face
- more than 7
- total of 13

sol:

i) $S = \{ \dots \}$ $n(S) = 36$

let

(i) A denotes sum of 8

$$A = \{(2,6), (3,5), (4,4), (5,3), (6,2)\}$$

$$n(A) = 5$$

$$P(A) = \frac{5}{36}$$

(ii) same no $B = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$

(iii) more than 7
 $P(C) = ?$

(iv) total of 13

Q) from an ordinary deck of playing cards, if a card is selected at random, what is probability that it is

(i) Red card

(ii) Diamond card

(iii) card of queen

(iv) card of no 4 & is a spade card

sol let A be card of red

(i) $P(A) = \frac{n(A)}{n(S)} = \frac{26}{52} = \frac{1}{2} = 0.5$

(ii) $P(B) = \frac{n(B)}{n(S)} = \frac{13}{52} = \frac{1}{4} = 0.25$

(iii) $P(C) = \frac{n(C)}{n(S)} = \frac{4 \cdot 48}{52 \cdot 51}$

Combination (ordered not matter)
Permutation (order matter)

$${}^n C_r$$

select, pick.

$n(S) = 52 C_1 = 52$

(iv) $P(D) = \frac{{}^1 C_1 \cdot {}^{51} C_0}{{}^{52} C_1} = \frac{1}{52}$

Book question (chap 6)

5.21 (a) $n(S) = 12$ total = 12
Bad = 2, Good = 10

Selected = 4 for making cake
 $n(S) = {}^{12} C_4 =$

(i) exactly one is bad:

let A ...
 $n(A) = {}^2 C_1 \cdot {}^{10} C_3$ (Multiplicative law)

$$P(A) = \frac{n(A)}{n(S)}$$

(ii) at least one is bad:

$$n(B) = ({}^2 C_1 \cdot {}^{10} C_3) + ({}^2 C_2 \cdot {}^{10} C_2)$$

$$P(B) = \frac{n(B)}{n(S)}$$

6.22) (a) $n(S) = 10$ ✓

(b) $n(S) = 20C_3$ (discrete values)

(1) 10 are even in first 20.

$$1^{st} + 2^{nd} + 3^{rd} = \text{even}$$

$$E + E + E = \text{Even}$$

$$E + O + O = \text{Even}$$

$$n(A) = ({}^{10}C_3 \cdot {}^{10}C_0) + ({}^{10}C_1 \cdot {}^{10}C_2)$$

$$P(A) = \frac{n(A)}{n(S)}$$

(i) Product is even:

$$E \times E \times E = E$$

$$E \times E \times O = E$$

$$E \times O \times O = E$$

$$n(B) = ({}^{10}C_3 \cdot {}^{10}C_0) + ({}^{10}C_2 \cdot {}^{10}C_1) + ({}^{10}C_1 \cdot {}^{10}C_2)$$

Stats Lec No 13 Friday
3-1-25
Independent events:
"Two events are said to be independent if the occurrence or non-occurrence of one event does not effect or change the occurrence or non-occurrence of other event."
Mathematically:
 $P(A \cap B) = P(A) \cdot P(B)$

→ for dependent events:
 $P(A \cap B) \neq P(A) \cdot P(B)$

* Mutually exclusive events:
"Two events are said to be mutually exclusive if one event occurs the other will not occur."
 $P(A \cup B) = P(A) + P(B)$

Additive model:
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $\therefore P(A \cap B) = P(A) \cdot P(B)$

for mutually exclusive
either $P(A) = 0$ or $P(B) = 0$

Exercise

6.27

* Hint: $O/R \rightarrow$ union ($\cup$)
 $AND \rightarrow$ intersection ($\cap$)

(b) Women = 20

Men = 10

Total = 30

Half men with brown eyes = 5
 Half women with brown eyes = 10

let $A \rightarrow$ Men $\rightarrow P(A|B) = P$
 and $B \rightarrow$ Brown eyes

As $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A) = \frac{10}{30}; P(B) = \frac{15}{30}; P(A \cap B) = \frac{5}{30}$$

from eq (1)

$$P(A \cup B) = \frac{10}{30} + \frac{15}{30} - \frac{5}{30} = \frac{20}{30}$$

$$P(A \cup B) = \frac{2}{3}$$

6.40) (i) $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$?

two dice sample space
 $S = \{ \dots \}$

$$E_1 = \{(1,6), (2,6), (3,6), (4,6), (6,6), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5)\}$$

$$n(E_1) = 11 \\ P(E_1) = \frac{11}{36}$$

$$E_2 = \{(1,5), (2,5), (3,5), (4,5), (5,5), (5,1), (5,2), (5,3), (5,4), (5,6)\}$$

$$n(E_2) = 10 \\ P(E_2) = \frac{10}{36}$$

$$P(E_1 \cap E_2) = \frac{11}{36} \times \frac{10}{36} = \frac{110}{1296}$$

$$E_1 \cap E_2 = \{(5,6), (6,5)\} \\ n(E_1 \cap E_2) = 2; P(E_1 \cap E_2) = \frac{2}{36}$$

$$P(E_1 \cap E_2) = \frac{11}{36} \times \frac{10}{36} = \frac{110}{1296}$$

(ii) E_2 & E_3 $[P(E_2 \cap E_3)]$?

$$E_3 = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$$

$$n(E_3) = 6 \\ P(E_3) = \frac{6}{36}$$

$$E_2 \cap E_3 = \{\emptyset\} \\ P(E_2 \cap E_3) = 0$$

$$P(E_2) \cdot P(E_3) = \frac{10}{36} \times \frac{6}{36} = \frac{60}{1296}$$

dependent
 (iii) $E_1 \cap E_3 = \{(6,6)\}$

$$P(E_1 \cap E_3) = \frac{1}{36}$$

$$P(E_1) \cdot P(E_3) = \frac{11}{36} \times \frac{6}{36} = \frac{66}{1296}$$

8-1-25

cc lec 14

Wednesday

(*) Conditional Probability:
 $\rightarrow P(A|B) \rightarrow$ find $P(A)$, given $P(A)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- Hint: given that (1) or
 already occurred or
 relative to the hypothesis or
 if

Q 6.34) (c)

let $A \rightarrow$ accounting Background
 let $B \rightarrow$ executive Background

$$P(A) = 0.20 \cong \frac{20}{100}$$

$$P(A \cap B) = 5\% = \frac{5}{100} = 0.05$$

$$P(B|A) = ?$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.05}{0.20} = \frac{1}{4}$$

6.51) Independent events

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

$$P(A) = 0.4, P(\bar{A}) = 1 - 0.4 = 0.6$$

$$P(B) = 0.5, P(\bar{B}) = 0.5$$

$$P(C) = 0.1, P(\bar{C}) = 0.9$$

(i) all ^{missiles} hit the target:
 let $D \rightarrow$ all missiles hit target

$$P(D) = P(A \cap B \cap C) \\
= P(A) \cdot P(B) \cdot P(C) \\
= 0.4 \times 0.5 \times 0.1 = 0.02$$

(ii) At least one of the three hit targets
 let $E \rightarrow$ at least " " "

$$P(E) = P(A \cap \bar{B} \cap \bar{C}) \cup P(\bar{A} \cap B \cap \bar{C}) \cup P(\bar{A} \cap \bar{B} \cap C) \cup P(A \cap B \cap \bar{C}) \cup P(A \cap \bar{B} \cap C) \cup P(\bar{A} \cap B \cap C) \cup P(A \cap B \cap C) \\
= P(A) \cdot P(\bar{B}) \cdot P(\bar{C}) + P(\bar{A}) \cdot P(B) \cdot P(\bar{C}) + \dots$$

complete it

(iii) exactly one

$$P(F) = P(A \cap \bar{B} \cap \bar{C}) \cup P(\bar{A} \cap B \cap \bar{C}) \cup P(\bar{A} \cap \bar{B} \cap C)$$

(iv) exactly two

$$P(G) = P(A \cap B \cap \bar{C}) \cup P(A \cap \bar{B} \cap C) \cup P(\bar{A} \cap B \cap C)$$

(*) Baye's theorem:

$$P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{\sum_{i=1}^k P(A_i) \cdot P(B|A_i)}$$

where it is used? { if $i=2$, 2 conditions }

in conditional prob, you have one condition
but in Bayes, you have more than one condition.

i) $A_1, A_2, A_3, \dots, A_k$ is called overall sample space

ii) $A_1 \cap B = A_2 \cap B = \dots = \emptyset$

$$P(A_i | B)$$



called posterior probabilities

$$\frac{P(A_i) P(B | A_i)}{\sum_{i=1}^k P(A_i) P(B | A_i)} \rightarrow \text{prior prob "pehle samajh jaoi"}$$

iii) sum of main prob = 1 $\leftarrow P(S)$

Ex 6.1)

$$P(A) = 0.50$$

$$P(B) = 0.30$$

$$P(C) = 0.20$$

let $D \rightarrow$ denotes fails to rise

$$P(D | A) = 0.02$$

$$P(D | B) = 0.03$$

$$P(D | C) = 0.05$$

$$\therefore P(A | D) = ?$$

$$P(A | D) = \frac{P(A) P(D | A)}{P(A) P(D | A) + P(B) P(D | B) + P(C) P(D | C)}$$

$$= \frac{0.50 \times 0.02}{0.50 \times 0.02 + 0.30 \times 0.03 + 0.20 \times 0.05}$$

$$P(D | A) = 6.34$$

Example

$$P(A) = 0.25,$$

$$P(B) = 0.35,$$

$$P(C) = 0.40$$

let $D \rightarrow$ defective

$$P(D | A) = 0.05$$

$$P(D | B) = 0.04$$

$$P(D | C) = 0.02$$

$$P(A | D) = \frac{P(A) \cdot P(D | A)}{P(A) \cdot P(D | A) + P(B) \cdot P(D | B) + P(C) \cdot P(D | C)} = ?$$

$$P(B | D) = ?$$

$$P(C | D) = ?$$

10-1-2025

STAT Lec 15 Friday

Random Variables

two types:

① Discrete random var.

① $f(x) \geq 0$

② $\sum f(x) = 1$
 $\nearrow$ sum of all prob

② Continuous random var:

① $f(x) \geq 0$

② $\int_{-\infty}^{\infty} f(x) dx = 1$
 $\downarrow$ indep var

$\hookrightarrow$ integration is used for continuous curve (can have point values)

$\therefore f(x) = P(\underbrace{X = x}_{\substack{\downarrow \\ \text{prob. random var}}})$ is Ki value

Example 7.1

Example of Discrete Random Variable

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$$n(S) = 8$$

Let "X" be random variable that denotes no. of heads

Then "X" can take values:-

$$X = 0, 1, 2, 3$$

$$f(0) = P(X=0) = \frac{1}{8}$$

$$f(1) = P(X=1) = \frac{3}{8}$$

$$f(2) = P(X=2) = \frac{3}{8}$$

$$f(3) = P(X=3) = \frac{1}{8}$$

• Probability Distribution for the No. of heads:

X	f(X)
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$
	$\frac{8}{8} = 1$

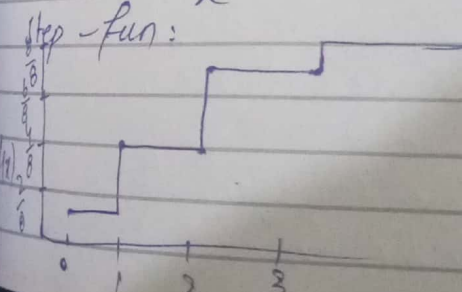
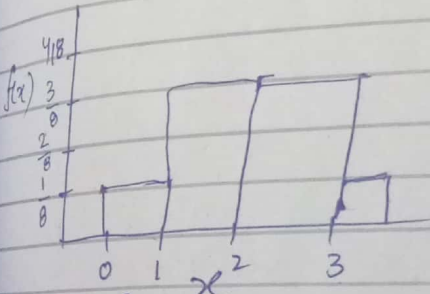
• Distribution function $F(x)$:-
cumulative distribution fun (akatha karna)
it is used when you have lot of prob
& you wanna see their prob.

compute C.D prob for this

if $x < 0$ $P(X < x) = 0$
 (b/c it is cumulative distribution)
 if $0 \leq x < 1$ $P(X \leq 1) = P(X=0) = \frac{1}{8}$
 if $1 \leq x < 2$ $P(X \leq 2) = P(X=0) + P(X=1) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$
 if $2 \leq x < 3$ $P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$
 for $x \geq 3$ $P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{8}{8} = 1$

its graph is called step - fun / step graph.

Histogram



Q) A bag contains 15 white balls and 7 black balls. If a ball is selected at random, write down the probability distribution that the balls selected is white ball.

White = 15
Black = 7
Total = 22 } Selected = 3

$$n(s) = {}^{22}C_3$$

Let "X" be random var that denotes white balls. Then "X" can take values

$$X = 0, 1, 2, 3$$

→ Prob Dist

X	P(X=x)
0	$\frac{{}^{15}C_0 \cdot {}^7C_3}{{}^{22}C_3} =$
1	$\frac{{}^{15}C_1 \cdot {}^7C_2}{{}^{22}C_3} =$
2	$\frac{{}^{15}C_2 \cdot {}^7C_1}{{}^{22}C_3} =$
3	$\frac{{}^{15}C_3 \cdot {}^7C_0}{{}^{22}C_3} =$

complete it.

Example 7.2 do it

→ Continuous Prob Distribution:

Example 7.3

$$f(x) = Kx \quad 0 \leq x \leq 2$$

$$= 0 \quad \text{elsewhere}$$

"continuous have limit in q"

(i) $f(x) \geq 0$

if $x=0, 1, 2, 3$ passed, for is ≥ 0

if this domain finished $f(x)=0$

$$P(X=3) = 0$$

(ii) $\int_{-\infty}^{+\infty} f(x) dx = 1$

$$\int_0^2 f(x) dx = 1$$

$$\int_0^2 Kx dx = 1$$

$$= K \left[\frac{x^2}{2} \right]_0^2 = 1$$

$$\Rightarrow K \left[\frac{2^2}{2} - \frac{0^2}{2} \right] = 1$$

$$\Rightarrow \frac{K}{2} = \frac{1}{2}$$

$$K = 1 \quad \boxed{f(x) = \frac{1}{2}x}$$

$$P(X > 1) = \int_1^2 f(x) dx$$

$$\int_1^2 \frac{1}{2}x dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} \right]_1^2$$

(this is one sample ans)

both (A)

$$f(x) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$$

find prob^{that} both of samples less than 1

$$P(X < 1) = \int_0^1 f(x) dx$$

if continuous ξ $P(X=1)=0$

$$\textcircled{\text{ii}} \quad F(x) = \int_0^x \text{ or } \int_{-\infty}^x f(x) dx$$

fun can only be made when you have var.